Assumption University
Vincent Mary School of Engineering, Science and Technology
Department of Computer Science

Course Outline

CSX3009 – Algorithm Design

Semester 2/2025

Course status: Major Required Course 3 credits
Pre-requisite: CSX3003 Data Structure and Algorithms

Class Meeting:	Section 541 (Lecture)	Time: 9:00 –12:00 Day: Tuesday Room: SLM0307
	Section 542 (Lecture)	Time: 9:00 –12:00 Day: Thursday Room: SLM0307

Instructor: Dr.Kwankamol Nongpong

Office: VMES0405 (Suvarnabhumi campus)
E-mail: kwankamolnng@au.edu

Office Hours: Friday 10:00-11:30
or schedule an appointment by email

Textbook:
- Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson and Ronald L. Rivest, The MIT Press, 3rd Edition
- Algorithms by Jeff Ericsson, An online book, http://www.cs.illinois.edu/~jeffe/teaching/algorithms/, Latest update December 25, 2013
- Algorithm Design., J. Kleinberg, E. Tardos, Addison Wesley, 2005.

References:
- Various online judge websites. For example, codeforces.com, acm.timus.ru, www.spoj.com, onlinejudge.u-aizu.ac.jp, dmoj.ca
- Python Algorithms, Magnus Lie Hetland, Apress, ISBN 978-1-4302-3237-7, 2010

Course Description:

Techniques for designing algorithms using divide and conquer, greedy method, dynamic programming and backtracking by emphasizing on analysis of efficiency, design techniques for NP problem domain.

Course Objectives:

This course would lead students through the challenges and current trends of natural language processing. Firstly, students will learn the basics of how words and sentences can be represented and standard text processing algorithms that are necessary for the downstream NLP tasks. The course will then give students a walk-through on some classic classifiers, neural networks and deep learning architectures and how they are applied in NLP field. Modern NLP tasks and applications on text and speech processing are also discussed in this course. Lastly, students will choose a downstream NLP task of their interest, implement algorithms/classifiers for such a task and present their techniques and findings to the class at the end of the semester.

Mark Allocation:

Assignments	20%
Midterm Examination	30%
Final Examination	30%
Term Project	20%
TOTAL	**100%**

Assignments:

Assignments shall be given to students from time to time to reinforce students' understanding of the rather theoretical materials of the course.

Class Schedule:

Week	541	542	Topics
1	Nov. 11	Nov. 13	Summary of the algorithm analysis and its correlation with the real program efficiency.
2	Nov. 18	Nov. 20	Python Rehearsal: • Iterative algorithms • Simple recursive algorithms
3	Nov. 25	Nov. 27	Recursion: A critical key in algorithm design
4	Dec. 2	Dec. 4	Memoization I: Introduction
5	Dec. 9	Dec. 11	Memoization II: 2-argument problems
6	Dec. 16	Dec. 18	Dynamic Programming I: Bottom-up
7	Dec. 23	*TBA*	Dynamic Programming II
Midterm Examination January 8, 2026 @ 12:00-14:00			
8	Dec. 30	Jan. 15	**Project Proposal Presentation**
9	Jan. 20	Jan. 22	Breadth-First Search
10	Jan. 27	Jan. 29	Uniform Cost Search
11	Feb. 3	Feb. 5	Heuristics Search
12	Feb. 10	Feb. 12	Backtracking & Pruning, Branch and Bound
13	Feb. 17	Feb. 19	Greedy Algorithm Divide and Conquer
14	Feb. 24	Feb. 26	**Term Project Presentation**
Final Examination March 10, 2026 @ 9:00-12:00			

**TBA* = To be announced

Project:

A group of no more than two members will study for an algorithmic solution for a specified problem (must be proposed to get approval by the lecturer) and implement the solution. The report will include the solution, its running time analysis, the implemented program, and the example executions, including the test cases. The completion of the project includes presentation.

Other Requirements:

1. 80% attendance is required.
2. 15 minutes late will be considered absent.
3. Students are not allowed to take the final exam in the case that they have been absent for 3 times or more.

AI Usage Policy

- Students may use AI tools (e.g., ChatGPT, GitHub Copilot) to support learning — for brainstorming, debugging, or clarifying concepts — but not to replace independent thinking.
- Any use of AI must be acknowledged in all submissions (e.g., "Used ChatGPT to explain recursion logic") and verified for accuracy.
- Copying or submitting AI-generated work without understanding or modification is considered academic misconduct.
- Students should be ready to explain their work or make small changes during oral checks.
- AI is a learning partner, not a shortcut — use it responsibly to enhance your understanding, not to bypass it.